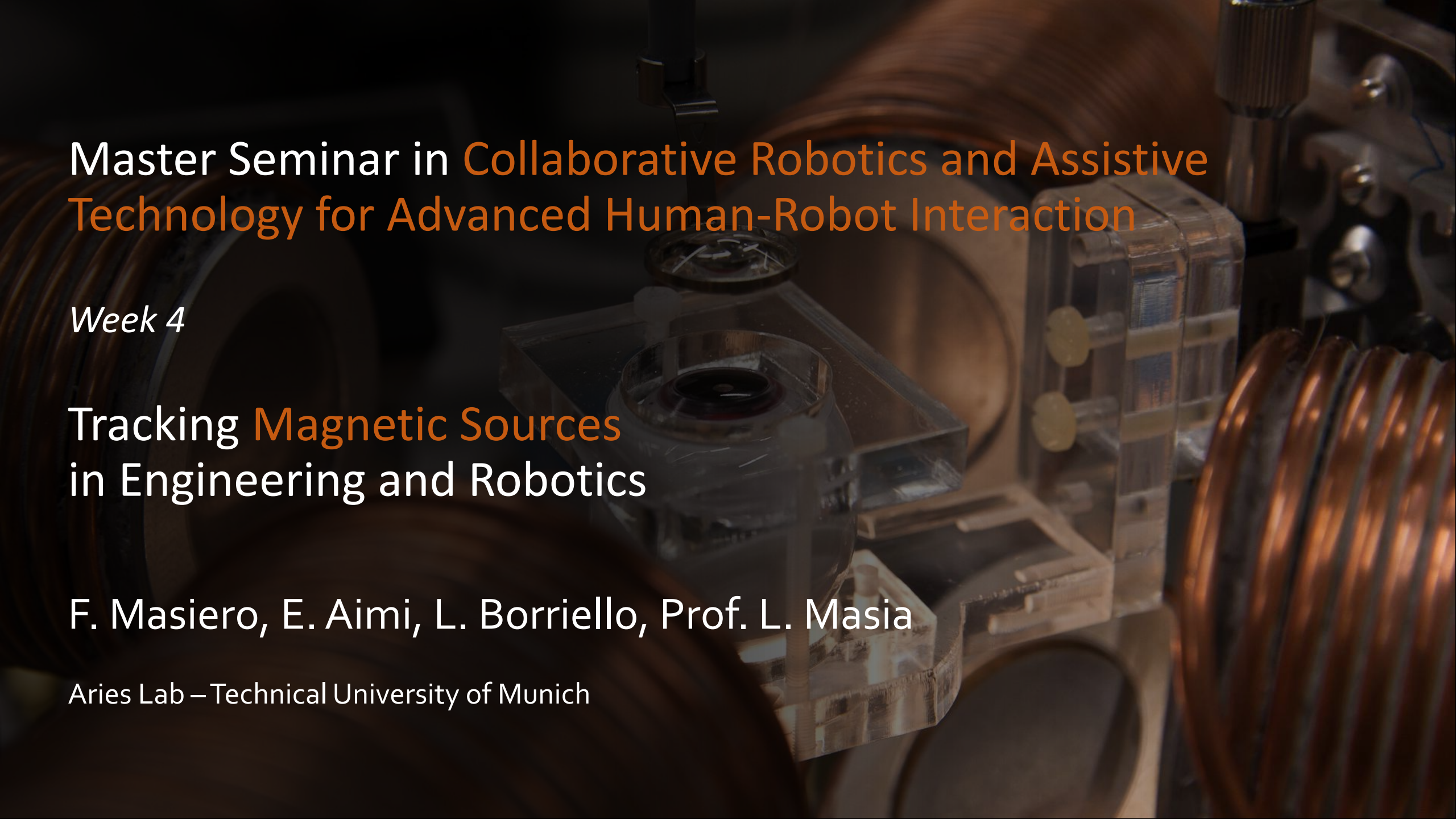




Master Seminar in Collaborative Robotics and Assistive Technology for Advanced Human-Robot Interaction

Week 4

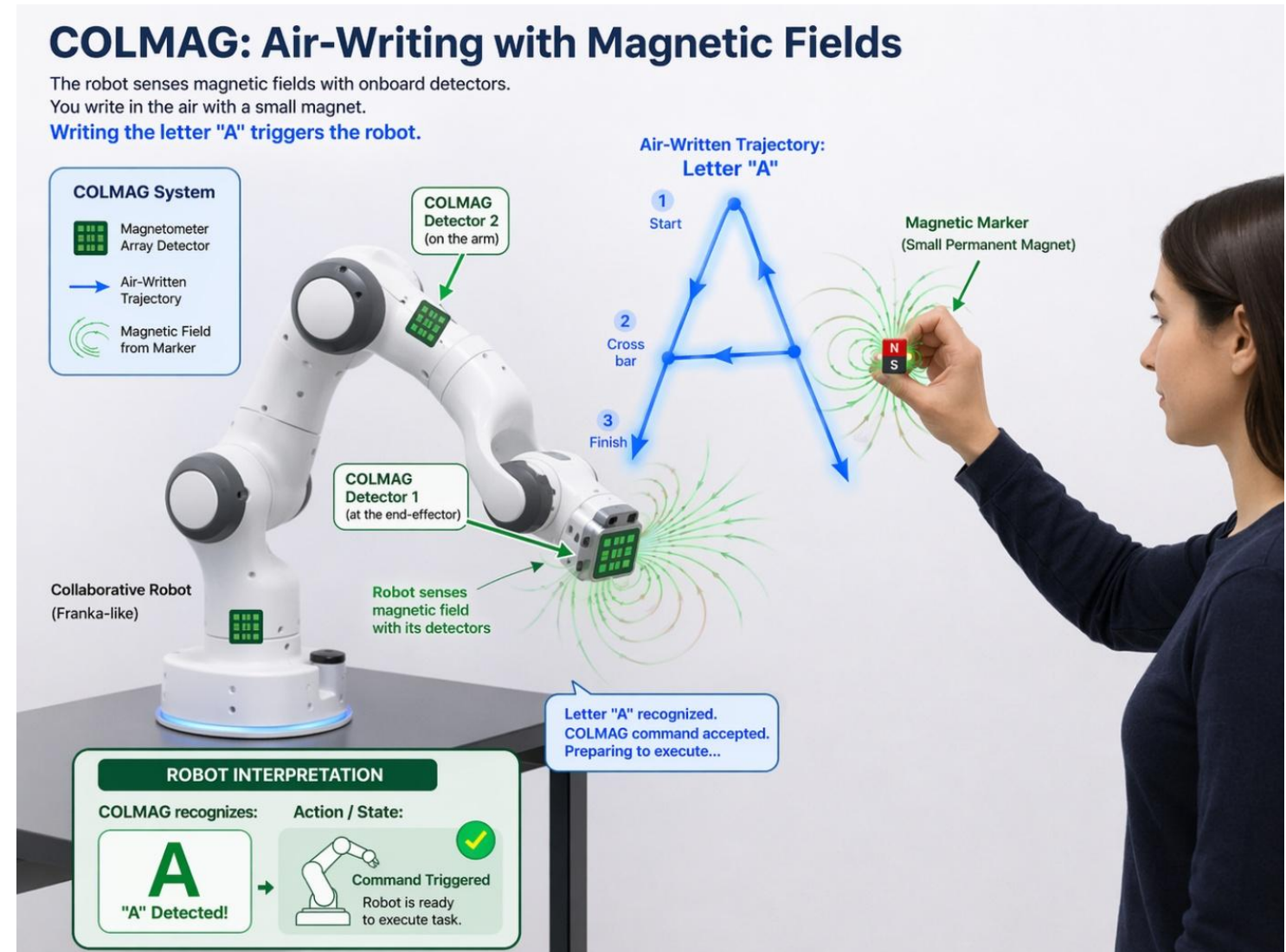
Tracking Magnetic Sources in Engineering and Robotics

F. Masiero, E. Aimi, L. Borriello, Prof. L. Masia

Aries Lab – Technical University of Munich

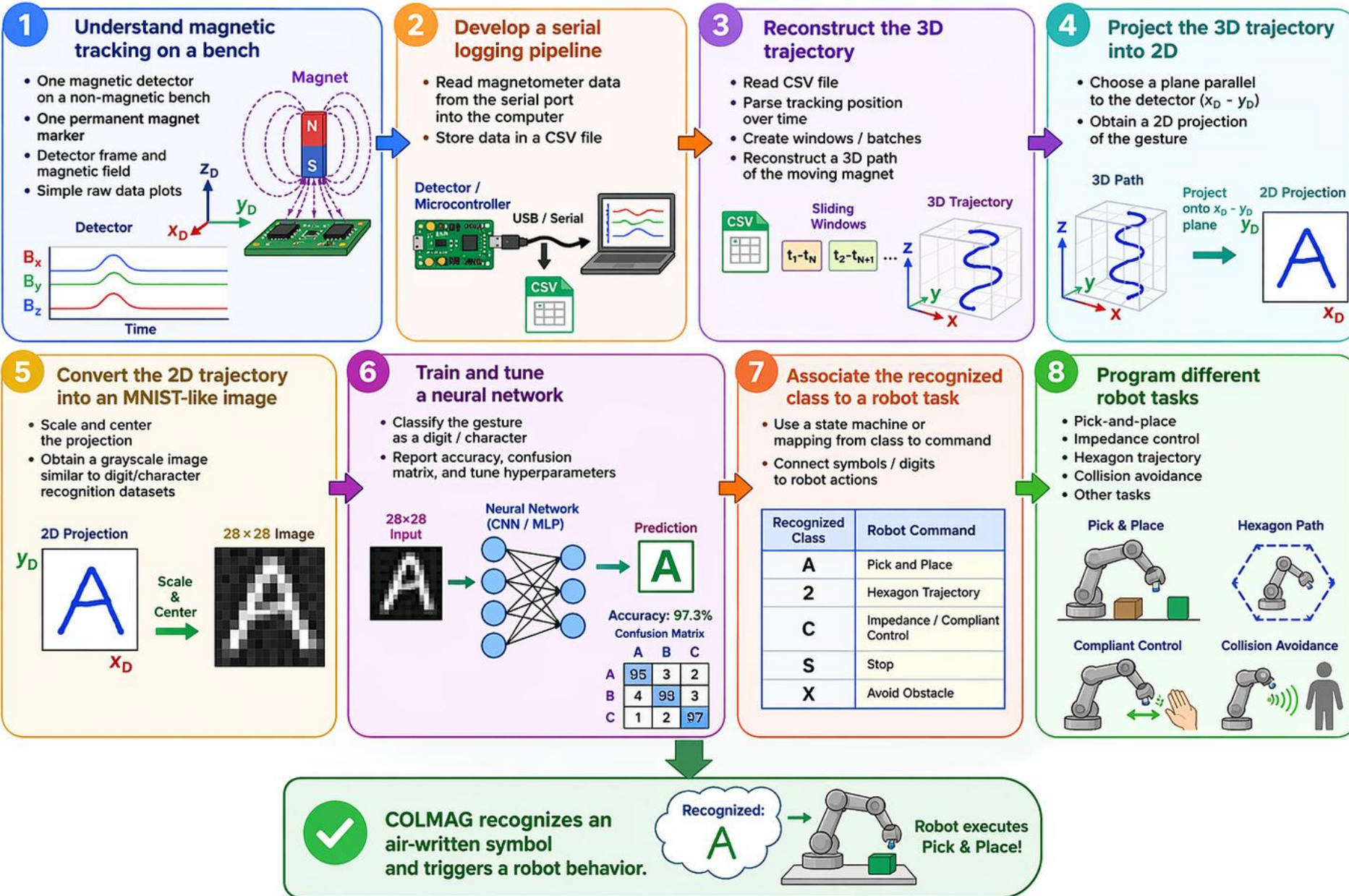
Projectual Work

Research project with an 8-week plan



COLMAG Work Plan

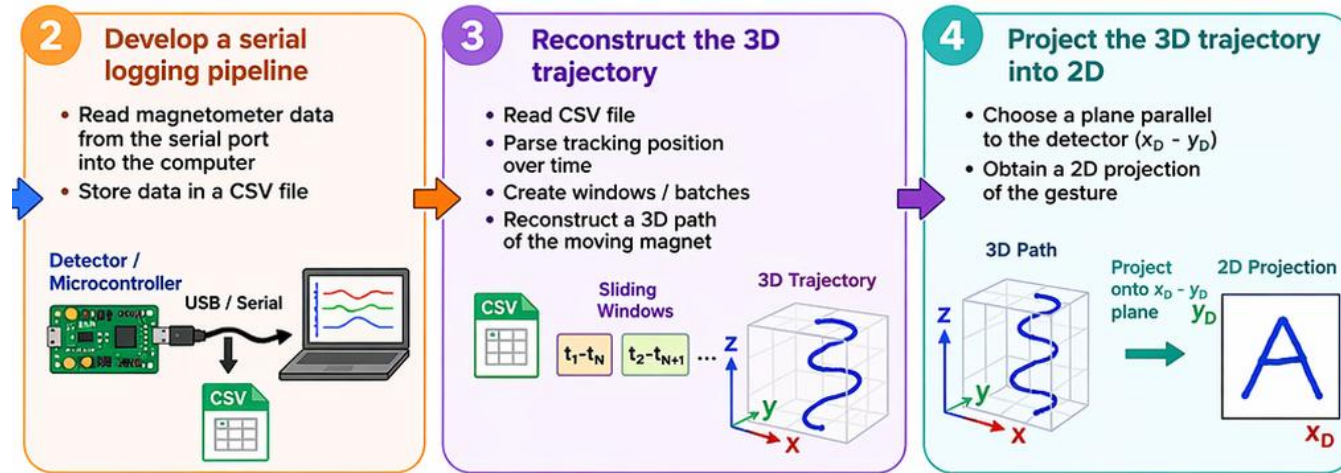
From magnetic sensing to robot actions



COLMAG Work Plan

From magnetic sensing to robot actions

In the last episode of the Seminar ...



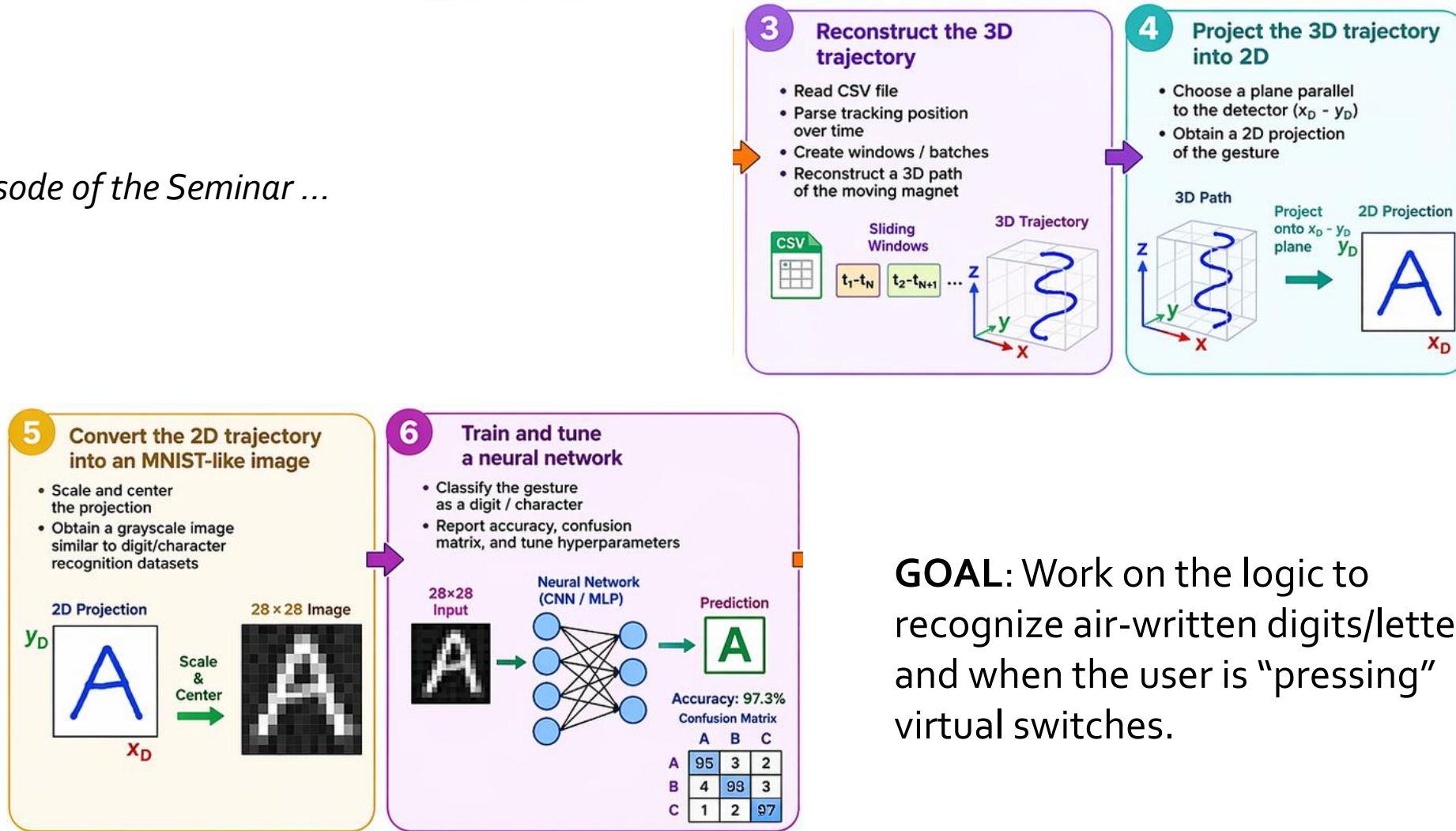
What you should have done, or what you should try to get by the end of the day

- *Plotting magnet position data in real-time for 3D visualization*
- *Project them in 2D to view digits*
- *Perform the pipeline to get NN-compatible images*
- *Define regions for virtual switches in the plane*

COLMAG Work Plan

From magnetic sensing to robot actions

In this episode of the Seminar ...



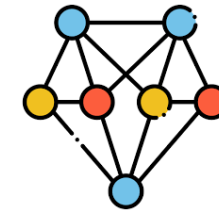
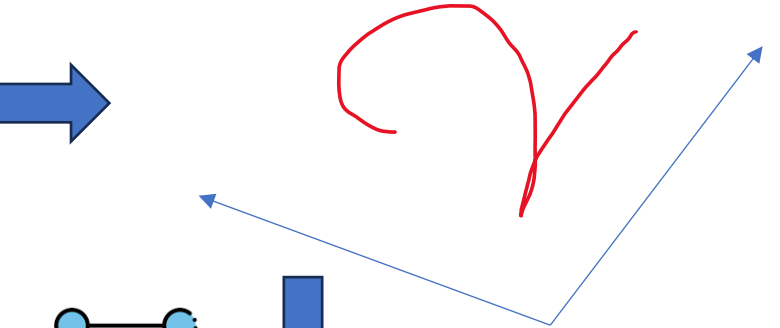
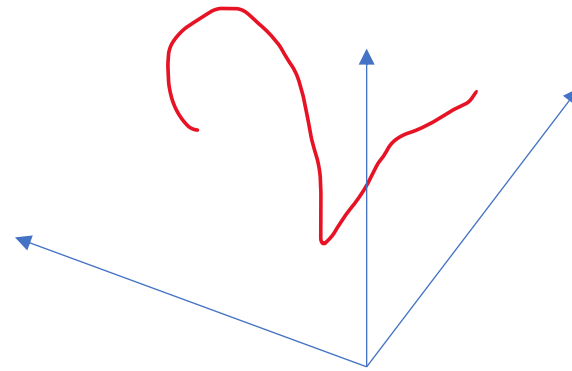
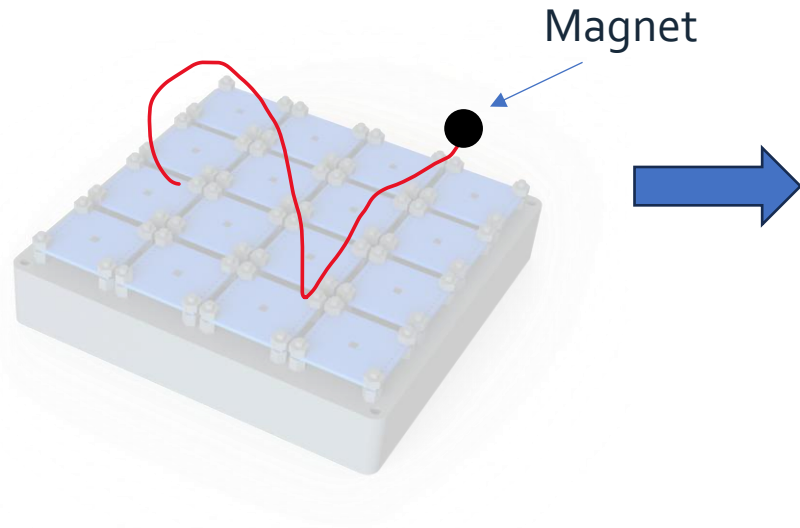
GOAL: Work on the logic to recognize air-written digits/letters and when the user is “pressing” virtual switches.

Projectual Work – Reminding you last Task

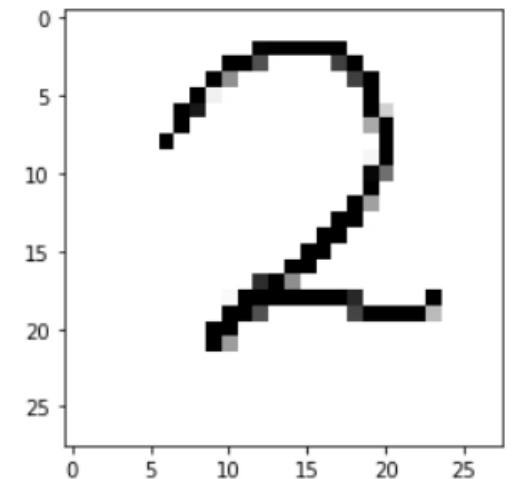
Real 3D Trajectory

3D representation in Python

2D representation in Python



*Get a 2D compressed image
representation of size
(N x N pixels)*

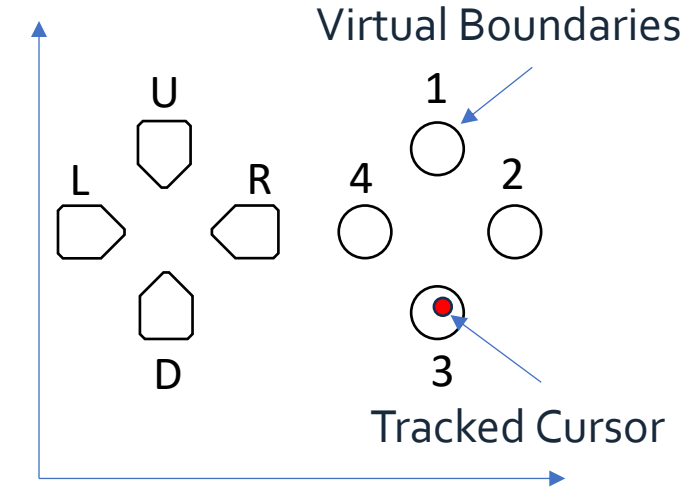
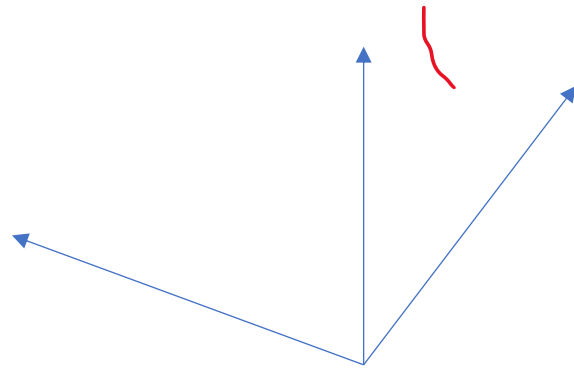
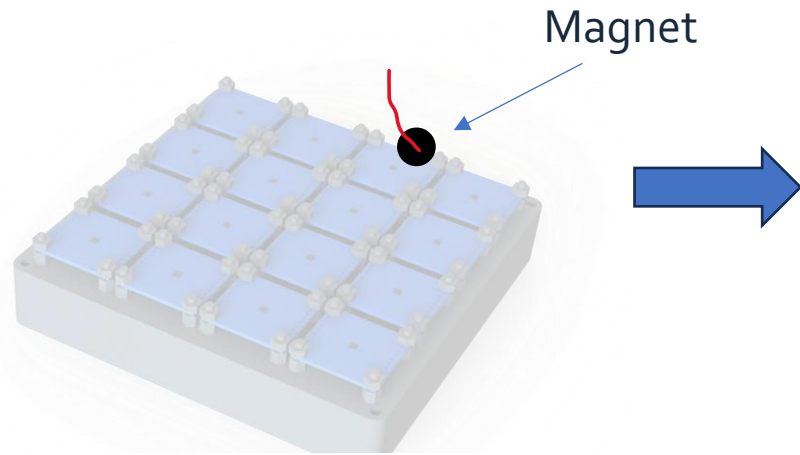


Projectual Work – Reminding you last task

Real 3D Trajectory

3D representation in Python

2D representation in Python

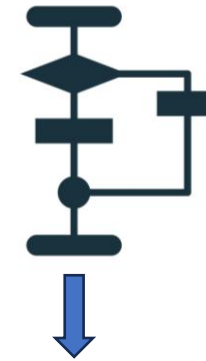


Create the Behavior of Virtual Switches



A suggested layout follows the one of joypads, but you can be creative.

Partition the projected space into multiple switches.



Create Logic to detect pressed virtual switches

"Pressed 3"

Projectual Work – Your Task

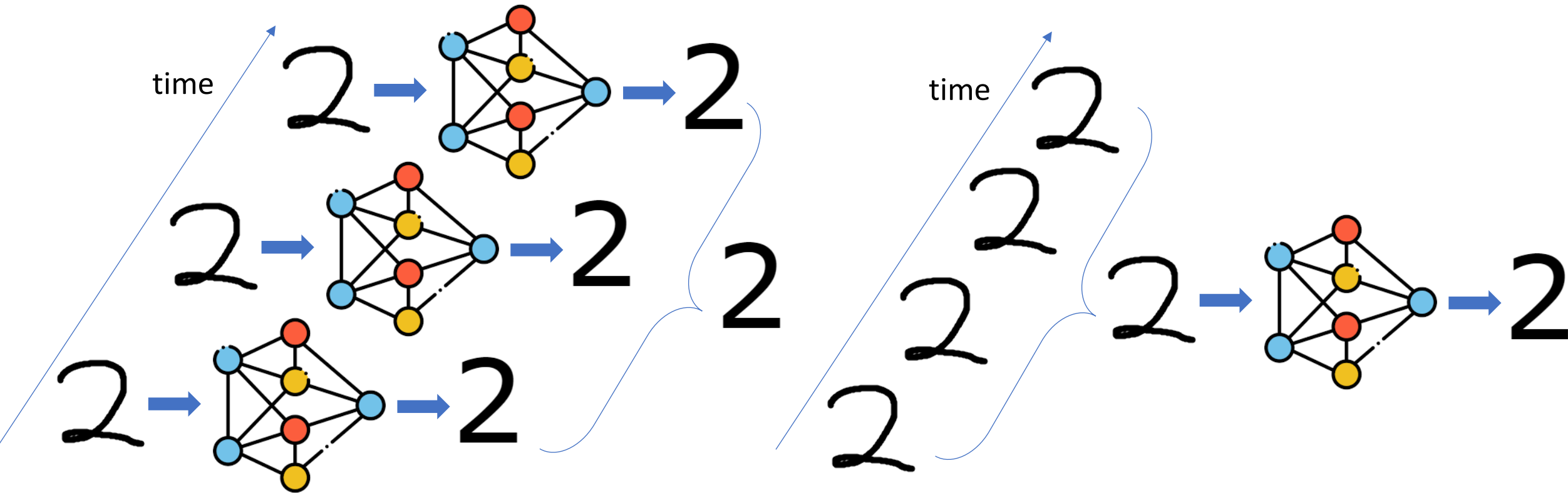
1. Understand how to efficiently process data for the project, *here are some suggestions*:
 - make sure **not to overload** your script too much **with visualization** (30 fps is sufficient).
 - make sure to **read data continuously**, at a fast speed (**important note**: don't read only 1 message in the serial port, read all the data available, store them and parse them to detect all correct messages).
2. Make detection more robust
 - flag **false tracking detection** and don't consider them (**important note**: when no magnet is on top of the detector, a random initialization is performed. Understand how you can get rid of all the detector false detections).
 - flag **false interactions**, i.e., random motion pattern on top of the detector (**important note**: you want to make sure to have a robust detection logic. For example, for digit recognition, you can use the confidence of the classification and measure if this is stable over a certain time window; another possibility could be to understand if the users are keeping writing or if they have finished).

Projectual Work – Your Task

Is it better to preprocess a lot the trajectory first and then give it to the NN for classification, or is it better to do minimal preprocessing, send almost raw data to the NN and the post-process it for detection?

https://cfml.se/blog/digit_recognition/

Think about this ...



Projectual Work – Your Task

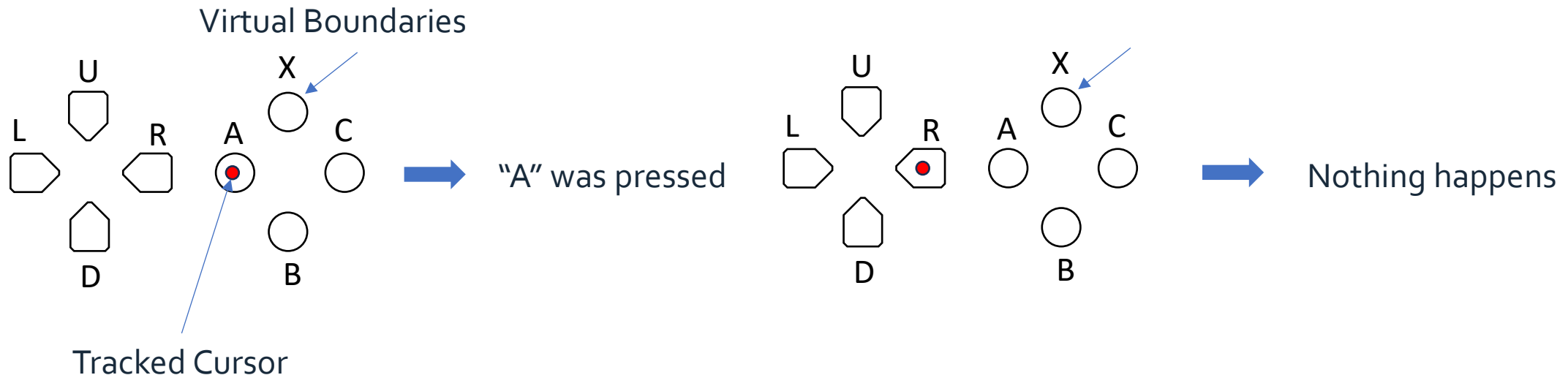
How can you process your data to make the recognition more robust:

- Temporal filtering (e.g., moving average with a certain window)
- Spatial filtering: after projection, you can use filters for greyscale images, which include denoising filters (smoothing), dilation or erosion (morphological operators), binarization, and many others. Note that also the **centering of the trajectory** matters!
- Analysis/check of the output: you can use statistical methods (confidence of the classification [softmax]), correlation of the initial image with some reference correct commands (correlation with a “1” ideal image).

Projectual Work – Your Task

3. Combine what you develop in a user interface:

- For example, I have a main menu with the switches, when I press the virtual arrow switches I get the corresponding output, but if I press A switch first, the arrows are blocked (meaning that A, block the motion commands of the robot for example). Another option could be, I press B and I start accepting digits command for few seconds (note that this helps making the interface robust 😊)

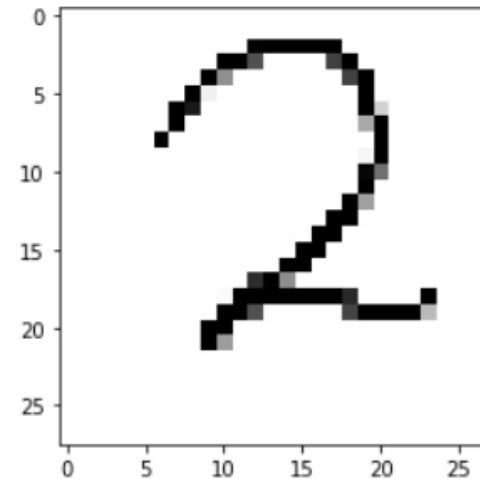
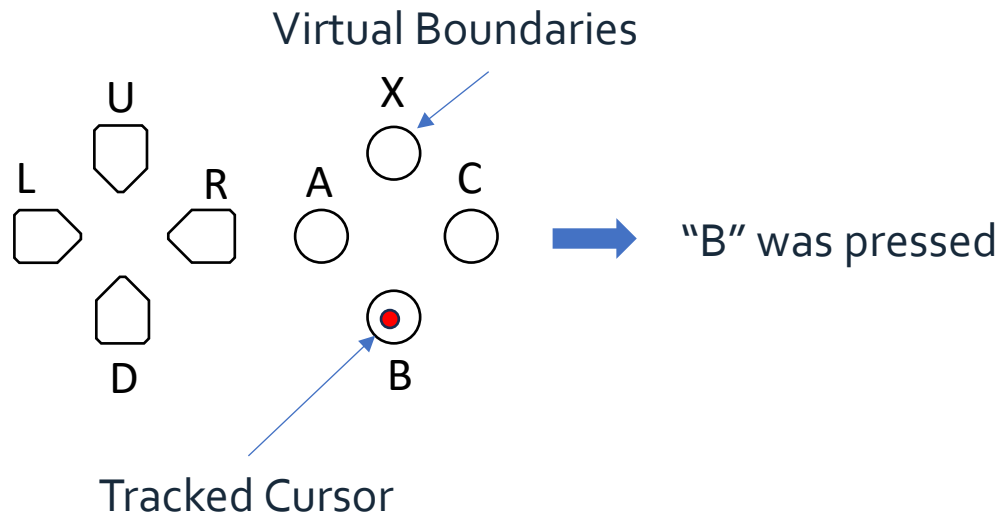


"A" act as a stop command for arrow keys

Projectual Work – Your Task

3. Combine what you develop in a user interface:

- For example, I have a main menu with the switches, when I press the virtual arrow switches I get the corresponding output, but if I press A switch first, the arrows are blocked (meaning that A, block the motion commands of the robot for example). Another option could be, I press B and I start accepting digits command for few seconds (note that this helps making the interface robust 😊)



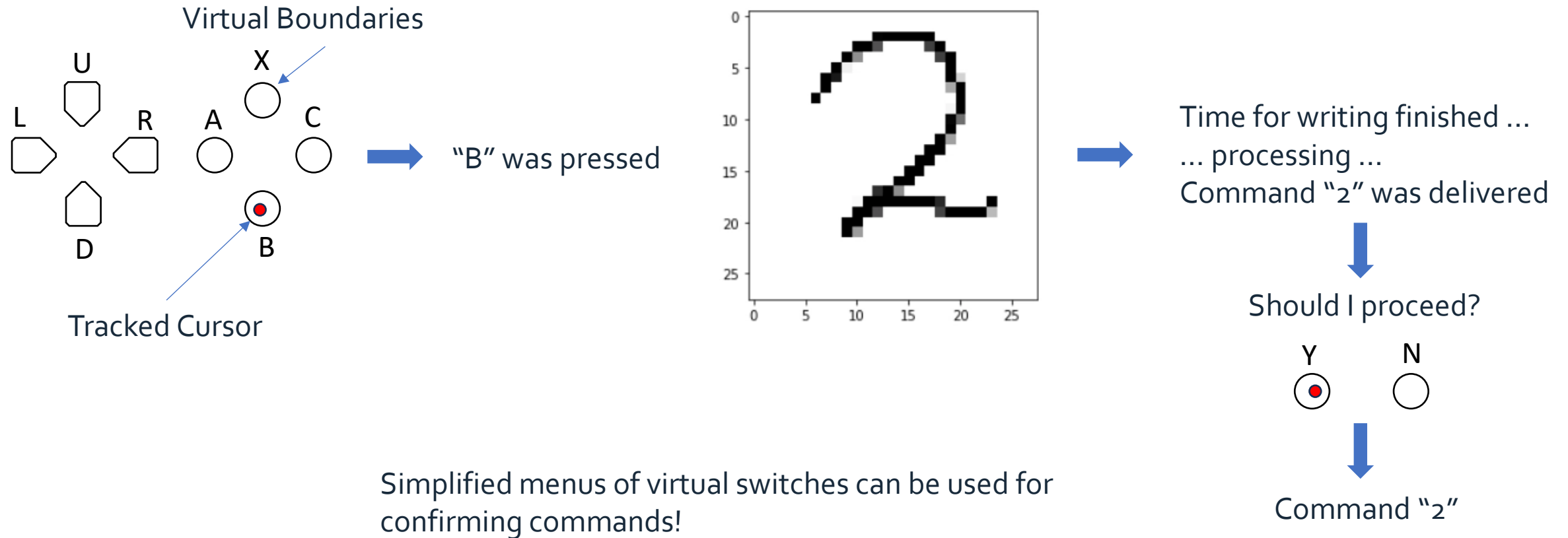
Time for writing finished ...
... processing ...
Command "2" was delivered

"B" enables for 3 seconds the digit recognition mode

Projectual Work – Your Task

3. Combine what you develop in a user interface:

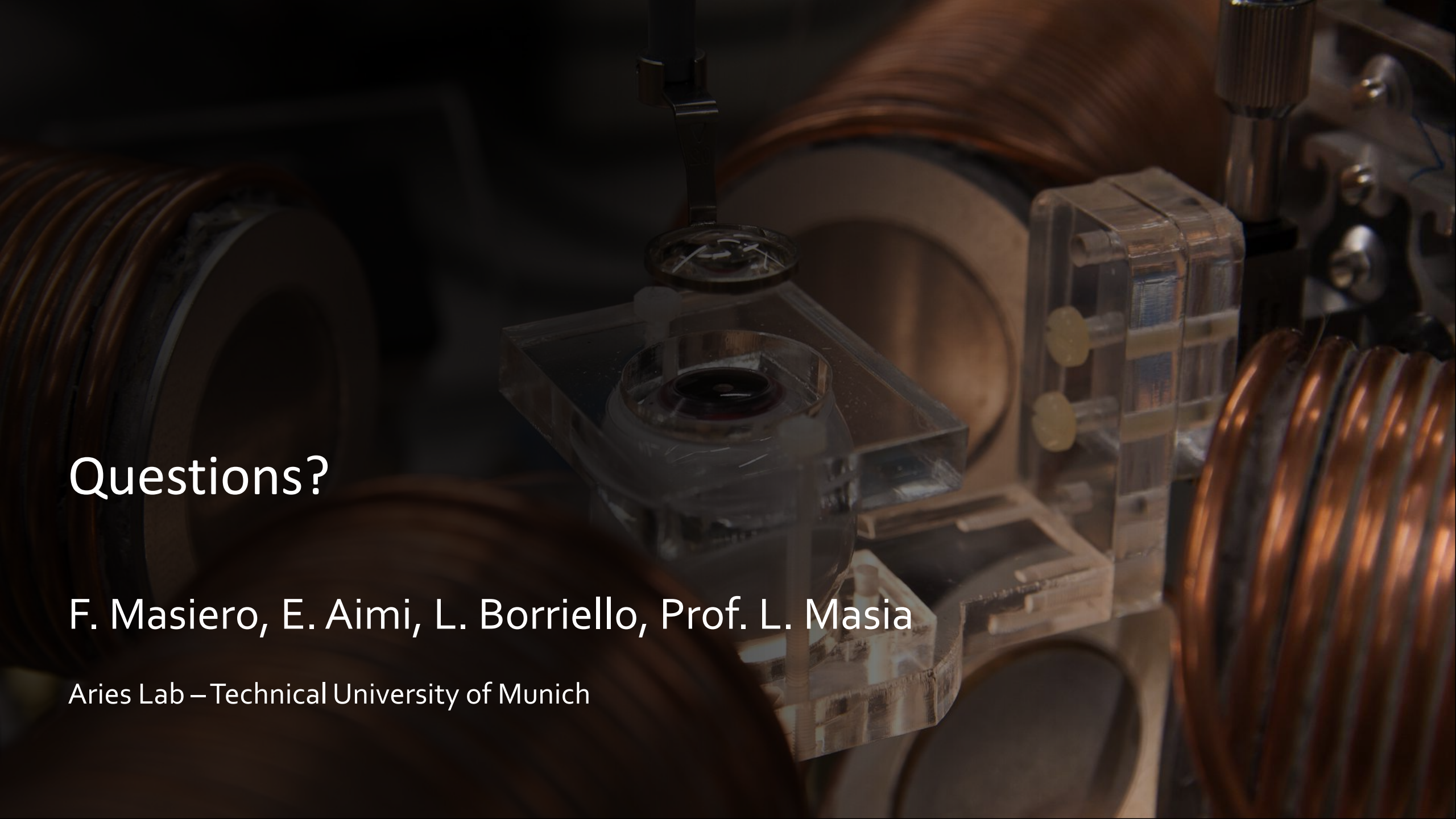
- You can also add logic for confirming important commands.



Projectual Work – Common Practice

THE HOLY RULES

- ☐ Do you think your solution is the best? Maybe not, discuss it with the others and with us; try to improve it!
- ☐ Keep your code tidy and well documented, not only in all the scripts you produce, but also in the repo Readme.MD
- ☐ Keep the repo clean and updated.



Questions?

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